Topological and Normed Linear Spaces

K. HÖFLINGER

It is our aim in this article to review those fundamental classes of topological linear spaces, in which the analysis of so-called "linear operators" is usually done. These spaces include, besides some generic classes like locally convex spaces and completely metrizable spaces, especially the class of normed linear spaces and related subclasses, where we want to emphasize Banach and Hilbert spaces.

1. Topological Linear Spaces

Basically, the scalar field $\mathbb{K}$ of the linear spaces under consideration will either be the field $\mathbb{R}$ of real numbers or the field $\mathbb{C}$ of complex numbers. Notice that these fields are furnished with the so-called standard topology, whose subbase is given by the totality of open intervals in the real line and by the totality of open disks in the complex plane, respectively. In general, if results on topological or normed linear spaces prove to be independent of the choice of the scalar field, we shall use the symbol $\mathbb{K}$ instead of $\mathbb{R}$ or $\mathbb{C}$.

By a *topological linear space* is meant an ordinary vector space V over a topological field $\mathbb{K}$ (usually that of real or complex numbers), together with a topology $\mathcal{T}_V$ such that addition $(x,y) \mapsto x+y$ and outer multiplication $(\lambda,x) \mapsto \lambda x$ are continuous. Hence a multiplicity of topological concepts, like the *closure* of a set, the *neighbourhood* of a point, the *convergence* of a sequence, the *continuity* of a mapping and others are meaningful in the context of topological linear spaces.

Let us continue with the introduction of a basic property that our topological linear spaces under consideration have to own: all singletons $\{x\}$ with $x \in V$ are closed sets, which is equivalent to say that the topology $\mathcal{T}_V$ fulfills HAUSDORFF's separation axiom. In other words, each pair of points $x, y \in X$ can be separated by open sets $\mathcal{O}_x, \mathcal{O}_y$, that is, there are $\mathcal{O}_x \subset V$ and $\mathcal{O}_y \subset V$ with $x \in \mathcal{O}_x$, $y \in \mathcal{O}_y$ and $\mathcal{O}_x \cap \mathcal{O}_y = \emptyset$. The validness of this separation axiom, in general topology also called the *T2-separation axiom*, implies not only that every singleton $\{x\}$ is closed, but also that the limit of a

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Email: *khoeflinger@t-online.de*, Url: https://khoefli.github.io/khMath.

convergent sequence of elements in V is unique.

Among a lot of properties, a topological linear space may have, there are two special ones that are of importance for our purposes:

1. A topological linear space V is said to be *metrizable*, if there is a function $d: V \times V \to [0, \infty)$ fulfilling the properties of a *metric*[1] such that the metric topology associated with d resembles the given topology $\mathcal{T}_V$.
2. A sequence $\{x_n\}_{n\in\mathbb{N}}$ with values in a topological linear space V is said to be Cauchy if for each neighbourhood $\mathcal{U}$ of $0 \in V$ there is an integer $N_\mathcal{U}$ such that $x_n - x_m \in \mathcal{U}$ for all $m, n > N_\mathcal{U}$. Then V is called *sequentially complete*, if every Cauchy sequence converges to a unique element $x \in V$. In the setting of metric spaces, however, the definition of Cauchy sequence may be simplified by means of the metric: $\{x_n\}_{n\in\mathbb{N}}$ is Cauchy, if for each $\epsilon > 0$ there is an integer N_ϵ such that $x_m - x_n \in B_\epsilon(0)$ for all $m, n > N_\epsilon$, where $B_\epsilon(0) := \{x \in V : d(x, 0) < \epsilon\}$ is the open ball with radius ϵ and center $0 \in V$.

For a topological linear space being metrizable and sequentially complete at the same time, both concepts of completeness are compatible to each other, that is, such a space is complete as a metric space if and only if it is sequentially complete as a topological linear space. This equivalence is founded by the fact that a metric space fulfills the *first countability axiom.*

In the sequel we also use the notion of *continuous embedding* of a topological linear space X into a topological linear space Y, which is nothing but a continuous and *injective* linear mapping $\iota: X \to Y$. The linearity of ι means that $\iota(\alpha x + \beta y) = \alpha\iota(x) + \beta\iota(y)$ holds for all $\alpha, \beta \in \mathbb{K}$ and $x, y \in X$.

In addition, if ι is surjective and ι^{-1} is continuous as well, then both mappings are called *linear homeomorphisms* between X and

[1] Given a non-empty set M, a mapping $d: M \times M \to \mathbb{R}$ is called a *metric* on M iff the following four conditions hold for all $x, y, z \in M$:

(M-1) $d(x, y) \geq 0$,
(M-2) $d(x, y) = 0$ if and only if $x = y$,
(M-3) $d(x, y) = d(y, x)$;
(M-4) $d(x, z) \leq d(x, y) + d(y, z)$ (*triangle inequality*).

Such a function d gives rise to a topology in M, called the *metric topology* associated with d, and a subbase is given by all open balls $B_r(x) := \{y \in M : d(x, y) < r\}$ with center $x \in M$ and radius $r > 0$.

Y. By $X \cong Y$ we mean that X and Y are linear-homeomorphic to each another.

2. Families of Seminorms

The class of *locally convex spaces* can be seen as the most important one within all kinds of topological linear spaces. The topology $\mathcal{T}_V$ of a locally convex topological linear space distinguishes by the fact that the origin 0 in V (and thus any other element) owns a neighbourhood base consisting of *convex*, *balanced* and *absorbing* sets.

It turns out that each such pair $(V, \mathcal{T}_V)$ gives rise to a family of *seminorms* on the space V. By the concept of seminorm we mean a mapping $|\cdot| : V \to \mathbb{R}$ having the properties

(SN-1) $|\lambda x| = |\lambda| \cdot |x|$ for $\lambda \in \mathbb{K}$ *(homogenity)*,

(SN-2) $|x + y| \leq |x| + |x|$ for $x, y \in V$ *(subadditivity)*,

which especially implies $|x| \leq 0$ for $x \in V$ and $|0| = 0$.

Moreover, if $\mathcal{U} \subset V$ forms a convex, balanced and absorbing neighbourhood of the origin 0, then the *Minkowski functional*

$$\begin{cases} |\cdot|_{\mathcal{U}} : V \to \mathbb{R}_0^+, \\ x \mapsto |x|_{\mathcal{U}} := \inf\{\, \lambda \geq 0 \,:\, x \in \lambda \mathcal{U} \,\} \end{cases}$$

fulfills all the properties of a seminorm. On the other hand, a family $\mathfrak{F}$ of seminorms defined on a linear space V leads in a natural way to a locally convex topology $\mathcal{T}_V$: it is the coarsest topology on V such that all seminorms in $\mathfrak{F}$ remain continuous.

In summary, there is a one-to-one correspondence between linear spaces endowed with locally convex topologies and those endowed with families of seminorms. Notice that a locally convex topology is Hausdorff iff the family of seminorms is *separating*, i. e. for each element $x \neq 0$ there is at least one seminorm $|\cdot|_\alpha$ such that $|x|_\alpha \neq 0$.

There are a two subclasses of locally convex spaces, which in the following we want to review:

1. A topological linear space V is called *F-space*, if it is also a *complete* metric space with translation-invariant metric d, i. e.
$$d(x, y) = d(x + z, y + z) \text{ for all } x, y, z \in V,$$
and the metric topology coincides with the given topology on V.
2. If the topology of the F-space V is locally convex, then V is said

to be a *Fréchet space*[2]. Alternatively, a locally convex space is Fréchet if and only if it is complete and its family of seminorms is mostly countable. In this case we may define a translation-invariant metric on a Fréchet space V according to

$$d(x,y) \;:=\; \sum_{n=1}^{\infty} p_n \cdot \frac{|x-y|_n}{1+|x-y|_n} \quad (x,y \in V)$$

where $\{p_n\}_{n=1}^{\infty}$ is any sequence of positive real numbers, whose infinite sum converges.

Given a topological $\mathbb{K}$-linear space X, let X^* be the set of all continuous linear mappings $\phi : X \to \mathbb{K}$, called the *adjoint space* X^* of X, which forms a linear space again. If the topology of X is at least locally convex, then a specific version of the celebrated HAHN-BANACH *theorem* ensures that there are non-trivial continuous linear mappings on X, thus we may assume $X^* \neq \{0\}$. We denote the elements of the adjoint space X^* as *continuous linear functionals.*

It is known from linear algebra that an ordinary $\mathbb{K}$-linear space V may be regarded as a linear subspace of its (algebraic) bi-adjoint space $V^{\#\#} := (V^{\#})^{\#}$, where $V^{\#}$ is the space of linear functionals $\phi : V \to \mathbb{K}$. More precisely, every element $x \in V$ gives rise to a linear functional $F_x : V^{\#} \to \mathbb{K}$ according to $F_x(\phi) := \phi(x)$ for all $\phi \in V^{\#}$. The natural mapping $\iota : V \to V^{\#\#}$, defined by $\iota(x) := F_x$ for $x \in V$, is linear and injective but in general not surjective.

Having these facts in mind, we introduce the *weak**-topology on the adjoint X^* of a locally convex topological linear space X: it is the smallest topology in X^* such that for each $x \in X$ the linear functional F_x remains continuous.

3. Normed Linear Spaces

Based on the results obtained in the last section, a normed linear space is nothing but a pre-Fréchet space, whose family of separating seminorms consits of only one element $\|\cdot\| := |\cdot|_0$, which of course must be a separating one, that is, $\|x\| = 0$ implies $x = 0$.

It is possible to introduce the concepts of norm and normed linear space in a direct way, i. e. without using the theory on families of seminorms and locally convex spaces. Following this approach, a

[2] A metrizable locally convex space that is not necessarily complete, is called a *pre-Fréchet space.*

normed space X is a $\mathbb{K}$-linear space that differs from other abstract linear spaces by the presence of a particular mapping $\|\cdot\| : X \to \mathbb{R}$, denoted the *norm* on X, which fulfills for all $x, y \in X$ and $\lambda \in \mathbb{K}$

(N-1) $\|\lambda x\| = |\lambda| \cdot \|x\|$,

(N-2) $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality)*,

(N-3) $\|x\| \geq 0$ and $\|x\| = 0 \Leftrightarrow x = 0$.

Linear spaces can be furnished with different norms. Then two arbitrary norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ defined on a linear space X are called *equivalent* to each another, if there are constants $c_1, c_2 > 0$ such that $c_1 \|x\|_\alpha \leq \|x\|_\beta \leq c_2 \|x\|_\alpha$ for all $x \in X$.

Each normed linear space X can be equipped with a very natural topology $\mathcal{T}$, called the *norm topology* on X. It can be defined either by the assertion that $\mathcal{T} := \mathcal{T}_{\|\cdot\|}$ is the smallest topology on X such that the norm mapping remains continuous, or it is defined to be the metric topology $\mathcal{T}_d$, when X is considered as a metric space with metric function $d(x, y) := \|x - y\|$ for all $x, y \in X$. In fact, both definitions lead to the same collection of open subsets in X, so $\mathcal{T} := \mathcal{T}_{\|\cdot\|} = \mathcal{T}_d$, which justifies to speak of *the* topology of a normed linear space. Clearly, equivalent norms on X own one and the same topology. For example, if X is a finite-dimensional linear space, then all norms on X are equivalent and each of them generates the so-called *standard topology* on X.

if X and Y are normed linear spaces and ι is a *norm-preserving* linear homeomorphism between X and Y, that is, $\|\iota x\|_Y = \|x\|_X$ for all $x \in X$, we denote this mapping a *norm-isomorphism* between X and Y, and in this case both spaces are called *norm-isomorphic* to each other (we again use the notion $X \cong Y$ in this case).

A normed linear space X is said to be *sequentially complete* (or briefly *complete*), if for any Cauchy sequence $\{x_n\}_{n\in\mathbb{N}}$ in X there is $x \in X$ such that $\lim_{n\to\infty} \|x_n - x\| = 0$. Equivalently, the space X is complete if and only if for every absolutely convergent series $\sum_{n=1}^{\infty} \|x_n\| < \infty$ there is $x \in X$ such that $\|x - x_n\| \to 0$ for $n \to \infty$. Each complete normed space is denoted a *Banach space*. Like in the non-complete case, a Banach space X with norm $\|\cdot\|$ is nothing but a Fréchet space, whose separating family of seminorms contains only this norm.

Notice that if the space X is normed, then $X^* = \mathscr{L}(X, \mathbb{K})$ is also normed, where the norm $\|\phi\|$ of a functional $\phi \in X^*$ is defined by

$$\|\phi\| := \sup_{x \in X,\, \|x\| \leq 1} |\phi(x)|.$$

Since the fields $\mathbb{R}$ and $\mathbb{C}$ are complete normed spaces as well, the adjoint space of a normed linear space is also complete. This follows from a basic theorem in functional analysis claiming that the space $\mathscr{L}(X, Y)$ of continuous linear mappings $\phi : X \to Y$ endowed with the *operator norm* is complete if and only if Y is complete.

In the following we want to introduce concepts of convergence for sequences in a *normed* linear space X. First of all, a sequence $\{x_n\}_{n \in \mathbb{N}}$ in X is called *strongly convergent* or simply *norm convergent* to $x \in X$, if it converges to x with respect to the norm topology, that is, if $\lim_{n \to \infty} \|x_n - x\| = 0$.

But there is a further and useful topology on X, called the *weak topology* $\mathcal{T}_w$. It is the coarsest topology on X such that all continuous linear functionals $\phi \in X^*$ remain continuous. The notion of convergence with respect to this topology is known as *weak convergence*. More precisely, a sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements in X is called *weakly convergent* to $x \in X$ (notation: $x_n \rightharpoonup x$), if $\lim_{n \to \infty} \phi(x_n) = \phi(x)$ for each $\phi \in X^*$.

The weak topology is also Hausdorff. But it does not contain more open sets than the norm topology, so $\mathcal{T}_w$ is weaker than $\mathcal{T}_{\|\cdot\|}$. In other words, norm convergence of any sequence in X implies weak convergence. If X is finite-dimensional, however, both topologies coincide, but this is in general not true in arbitrary normed linear spaces.

4. Special Properties of Banach Spaces

In this section we briefly review two fundamental properties, a Banach space X may or may not have. The first topic is *uniform* and *strict convexity* of real Banach spaces, while the second one is *reflexivity*.

A *uniformly convex space* is a *real* normed space X having the following property: for every $\epsilon > 0$ there is $\delta > 0$ such that for any elements x, y of the closed unit ball $B_1(0)$ of X the validness of the inequality $\|(x + y)/2\| > 1 - \delta$ implies $\|x - y\| < \epsilon$. In other words, if the center $(x + y)/2$ of the line segment $[x, y]$ between elements x, y belonging to the *unit sphere* $S_1(0) := \partial B_1(0)$ tends to $S_1(0)$, then the x and y tend together. This property is possibly lost when switching to another but equivalent norm on X.

A normed linear space X is called *strictly convex*, if the conditions $\|x\| \leq 1$, $\|y\| \leq 1$ and $x \neq y$ imply $\|x+y\|/2 < 1$. This means, that the *inner* line segment (x, y) between $x, y \in S_1(0)$ lies completely in the interior of $B_1(0)$, that is $\|\alpha x + (1-\alpha)y\| < 1$ for $0 < \alpha < 1$. Each linear subspace of a uniformly convex Banach space owns this feature. In the light of the forthcoming theorem 4.1, it is an important fact that uniformly convex spaces are also strictly convex.

Recall that a subset $\mathcal{C}$ of a linear space is called *convex*, if the line segment $[x, y]$ of any pair of points $x, y \in \mathcal{C}$ belongs to $\mathcal{C}$ again. Then many approximation problems may be formulated as follows: it is to find an element y_0 in a convex set $\mathcal{C} \subset X$ such that y_0 has minimal distance to a given element $y \notin \mathcal{C}$. The following statement, known as the *approximation theorem*, provides sufficient conditions regarding existence and uniqueness of a solution for that problem:

THEOREM 4.1. Let $(X, \|\cdot\|)$ be a real normed linear space, let $y \in X$ and $\mathcal{C} \subset X$ be a *closed* and *convex* subset. Then we have:

(a) *Uniqueness*: if X is *strictly convex*, then there is mostly one element $y_0 \in \mathcal{C}$ such that $\|y_0 - y\| = \inf_{x \in \mathcal{C}} \|y - x\|$.

(b) *Existence*: if X is a *uniformly convex Banach space* then there is a unique element $y_0 \in V$ such that $\|y_0 - y\| = \inf_{x \in \mathcal{C}} \|y - x\|$.

We next introduce the notion of reflexivity of Banach spaces. It may happen that for a given normed linear space X the natural embedding $\iota : X \hookrightarrow X^{**}$ is surjective, and in this case it constitutes a norm-isomorphism between X and X^*. Many normed linear spaces considered in functional analysis have this remarkable property, which is indicated by the concept of *reflexivity*: the space X is called *reflexive*, if ι is onto and consequently $X \cong X^{**}$.

We collect some propositions concerning reflexivity:

1. Since adjoint spaces of normed spaces are throughout Banach spaces, every reflexive space must be complete.
2. If a Banach space X is norm-isomorphic to a reflexive space, then X must be reflexive, too.
3. Dual spaces of reflexive spaces are reflexive.
4. Reflexivity of a Banach space X is strongly related to reflexivity of *all* its adjoint spaces, that is, X is reflexive iff one of the spaces $X^*, X^{**}, X^{***}, \ldots$ is reflexive. On the other hand, if X is not reflexive, then all these spaces prove to be

pairwise unisomorphic to each other.

Important examples of reflexive Banach spaces are provided by the MILMAN-PETTIS *theorem*: each *uniformly convex Banach space* proves to be reflexive.

5. Hilbert Spaces

Given a real or complex linear space H, an *inner product* in H is defined to be a mapping $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{K}$, which is

(I-1) linear in the first argument,

(I-2) either linear or antilinear in the second argument (conditional upon, whether $\mathbb{K}$ is the real or the complex number field),

together with the additional properties

(I-3) $\langle y, x \rangle = \langle x, y \rangle$ ($\mathbb{K} = \mathbb{R}$) rsp. $\langle y, x \rangle = \overline{\langle x, y \rangle}$ ($\mathbb{K} = \mathbb{C}$),

(I-4) $\langle x, x \rangle \geq 0$ for all $x \neq 0$,

(I-5) $\langle x, x \rangle = 0$ implies $x = 0$.

Property (I-3) is denoted the *Hermitean symmetry* of $\langle \cdot, \cdot \rangle$, while (I-4) and (I-5) together mean *positive definiteness*. The presence of the inner product gives rise to define a norm in H according to

$$\|x\| \;:=\; \langle x, x \rangle^{1/2} \qquad (x \in H), \tag{5.1}$$

so each linear space furnished with an inner product is normed in a natural manner. The inner product and the norm are coupled by the *Cauchy-Schwarz inequality*

$$|\langle x, y \rangle| \;\leq\; \|x\| \cdot \|y\| \qquad (x, y \in H),$$

which implies that the mapping $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{K}$ is continuous. An inner product space H is called a *Hilbert space* or shortly *Hilbertian*, if it is complete with respect to the norm defined by (5.1).

The pleasant geometric properties of inner product spaces may be attributed to the validness of the *parallelogram equation*

$$2\,\|x\|^2 + 2\,\|y\|^2 \;=\; \|x + y\|^2 + \|x - y\|^2 \qquad \text{for all } x, y \in H. \tag{5.2}$$

In general, the norm $\|\cdot\|$ on a linear space X can be induced by an inner product if and only if (5.2) holds. In this case and if X is a complex linear space, the inner product can be recovered from the norm by the *polarization formula*

$$\langle x, y \rangle \;=\; \frac{1}{4}\,(\,\|x + y\|^2 - \|x - y\|^2 + i\,\|x + iy\|^2 - i\,\|x - iy\|^2\,).$$

If X is a real space, however, the polarization formula simplifies to

$$\langle x, y\rangle \;=\; \frac{1}{4}\,(\,\|x+y\|^2 - \|x-y\|^2\,).$$

In the following we want to point out that Hilbert spaces differ from general Banach spaces by a couple of desirable properties:

1. *Orthogonality.* Two elements $x, y \in H$ are called *orthogonal* to each other, if $\langle x, y\rangle = 0$ (notation: $x \perp y$). Any subset $\mathfrak{O} \subset H$ is said to be *orthogonal* or an *orthogonal system*, if all elements in $\mathfrak{O}$ are orthogonal to each other. Finally, the *orthogonal space* $\mathcal{A}^\perp$ of a subset $\mathcal{A} \subset H$ contains exactly the elements $y \in H$, which are orthogonal to all $x \in \mathcal{A}$. Such sets basically form closed linear subspaces of H.

2. *Projection Theorem.* Let $\mathcal{A} \subset H$ be a *closed* linear subspace. Then for each $x \in H$ there is exactly one $x_0 \in \mathcal{A}$ such that

   $$\|x - x_0\| \;=\; \inf\{\,\|x-y\| : y \in \mathcal{A}\,\}.$$

   The element x_0 is called the *projection* of x onto $\mathcal{A}$ and minimizes the distance between $\mathcal{A}$ and x. The associated linear mapping $\pi_{\mathcal{A}} : x \in H \mapsto x_0 \in \mathcal{A}$ is also denoted the *projection* of H onto $\mathcal{A}$. In general, by a *projection* is meant any linear mapping $\pi : H \to H$ satisfying $\pi \circ \pi = \pi$ and $\langle \pi x, y\rangle = \langle x, \pi y\rangle$ for all $x, y \in H$.

3. *Decomposition Theorem.* Using the notation of orthogonal space $\mathcal{A}^\perp$, the projection of x onto $\mathcal{A}$ is characterized as follows: $x_0 = \pi_{\mathcal{A}} x$ iff $x - x_0 \in \mathcal{A}^\perp$. Hence each $x \in H$ can be uniquely represented by the formula $x = x_0 + x^\perp$, where $x_0 \in \mathcal{A}$ and $x^\perp \in \mathcal{A}^\perp$, so the entire space H is isometrically isomorphic to the direct sum $\mathcal{A} \oplus \mathcal{A}^\perp$ of the closed subspaces $\mathcal{A}$ and $\mathcal{A}^\perp$. This fact is known as *decomposition theorem* in Hilbert spaces. In other words, each closed linear subspace $\mathcal{A}$ of H is *topologically complementable*, that is, to each closed subspace $\mathcal{A}$ there is a second closed subspace $\mathcal{B} \subset H$ such that $\mathcal{A} \cap \mathcal{B} = \{0\}$ and $[\mathcal{A} \cup \mathcal{B}] = H$. Of course, a closed subspace $\mathcal{A}$ is topologically complemented by its orthogonal space.

4. *Existence of Orthonormal Bases.* An orthogonal system $\mathfrak{O} \subset H$ is denoted an *orthonormal system*, if $\langle x, x\rangle = 1$ for all $x \in \mathfrak{O}$. However, if $\mathfrak{O}$ contains only *finitely* many elements, then the span $[\mathfrak{O}]$ forms a *closed* subspace in H, and the projection $\pi_{[\mathfrak{O}]}$ of any element x onto $[\mathfrak{O}]$ can be expressed by the formula

$$\pi_{[\mathfrak{O}]}\, x \;=\; \sum_{i=1}^{n} \langle x, e_i\rangle\, e_i.$$

However, if $\mathfrak{O}$ consists of arbitrarily many elements, the inequality $\langle x, e_\alpha\rangle \neq 0$ can only be valid for mostly countable many elements $e_\alpha \in \mathfrak{O}$, which in conclusion enables to add the coefficients $\langle x, e_\alpha\rangle$ together about all elements of $\mathfrak{O}$.

An orthonormal system $\mathfrak{O}$ is called an *orthonormal base* in H, if its span $[\mathfrak{O}]$ lies dense in H. In this case we may represent each $x \in H$ according to

$$x \;=\; \sum_{e_\alpha \in \mathfrak{O}} \langle x, e_\alpha\rangle\, e_\alpha, \tag{5.3}$$

which is primarily a formal sum, but consists of mostly countable many summands. The inner products $\langle x, e_\alpha\rangle$ are called the *Fourier coefficients* of x with respect to $\mathfrak{O}$. The right hand side of (5.3) consists of *mostly countable many* non-zero summands, and this series is invariant by realignments of the elements $e_\alpha \in \mathfrak{O}$.

There is of course the need to determine, whether a given orthonormal system $\mathfrak{O}$ forms an orthonormal base. In fact, it can be shown that the following statements are equivalent:

(a) $\mathfrak{O}$ is an orthonormal base in H.

(b) $\mathfrak{O}$ is *complete*, that is, $\langle x, e\rangle = 0$ for all $e \in \mathfrak{O}$ implies $x = 0$.

(c) $\sum_{e_\alpha \in \mathfrak{O}} |\langle x, e_\alpha\rangle|^2 = \|x\|^2$ for all $x \in H$ (*Parseval's equation*).

It is also nearby to ask for the existence of orthonormal bases. First of all, the collection of all orthonormal systems in a Hilbert space H forms a partial ordered set, and under the assumption that ZORN's lemma[3] is valid, it contains a *maximal* element $\mathfrak{O}$ that cannot be increased, hence each Hilbert space $H \neq \{0\}$ owns an orthonormal base. Furthermore, if $\mathfrak{O}_1$ and $\mathfrak{O}_2$ are orthonormal bases in H, then there is a bijective mapping between them, thus they own identical cardinal numbers. This legitimates us to introduce the notation of *Hilbert space dimension* $|H|$ being the cardinal number of any orthonormal base in H.

A Hilbert space H is called *separable* if there is a mostly countable subset lying dense in H. It turns out that H is separable

[3] The *axion of choice* claims that if $\mathcal{F}$ is a disjoint family F_α ($\alpha \in I$) of non-empty sets, then there is a set A that shares exactly one element with each other set $F_\alpha \in \mathcal{F}$.

if and only if $|H| \leq |\mathbb{N}|$. Indeed, each orthonormal base in H forms a mostly countable set in this case, and H proves to be norm-isomorphic to the Hilbert space ℓ^2 of square summarizable sequences $f : \mathbb{N} \to \mathbb{K}$.

The following figure shows the inheritance tree of topological and normed linear spaces, where Banach and Hilbert spaces are positioned at the very end of this hierarchy:

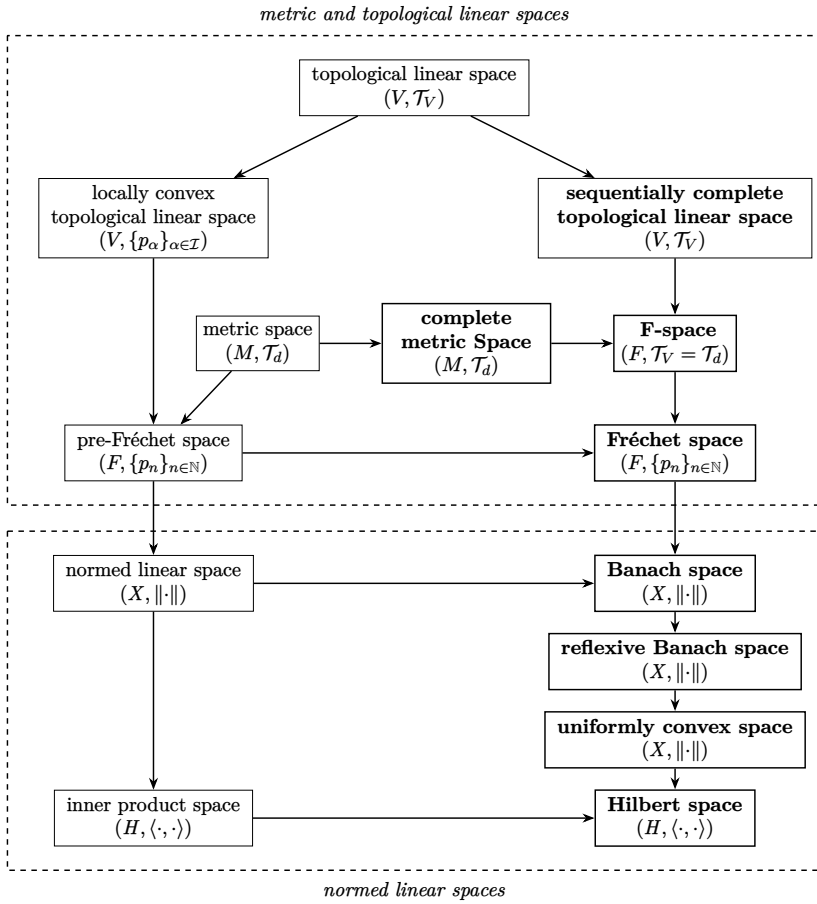


Figure 5.1: The hierarchy of metric and topological linear spaces

6. Bochner-Sobolev Spaces

This section is preliminary in order to outline the variational approach to evolution equations in the Hilbert space setting.

The *Bochner spaces* $L^p(\Omega, X)$ are nothing but Lebesgue spaces for vector-valued functions $f : \Omega \to X$, where X is a separable Banach space. To define them, we first have to introduce a concept of measurability and integration for vector-valued functions, that forms a generalization of the ordinary Lebesgue integral.

A function $f : \Omega \to X$ is called *weakly measurable*, if $\phi \circ f : \Omega \to \mathbb{K}$ is measurable for every $\phi \in X^*$. Then a weakly measurable function $f : \Omega \to X$ is called *Bochner-integrable*, if

$$\int_\Omega \|f\| \, d\mu < \infty.$$

For any $p \geq 1$ we denote $L^p(\Omega, X)$ the space of functions $f : \Omega \to X$ having Bochner-integrable powers $|f|^p$. The elements of this space have also to be understood as equivalence classes of functions being identical almost everywhere in Ω. Endowed with the norm

$$f \mapsto \|f\|_p := (\int_\Omega \|f\|^p \, d\mu)^{1/p},$$

the Bochner spaces $L^p(\Omega, X)$ are complete and thus constitute Banach spaces.

It should be mentioned that if we suppose the Banach space X to be separable, then a function $f : \Omega \to X$ is weakly measurable if and only if f is strongly (or Pettis-) measurable. The latter notion of measurability uses approximations of f by sequences $\{e_n\}_{n \in \mathbb{N}}$ of elementary functions $e_n : \Omega \to X$, and the Bochner integral of f is obtained by taking the limit of the Bochner integrals of the functions e_n, that is,

$$\int_\Omega f \, d\mu := \lim_{n \to \infty} \int_\Omega e_n \, d\mu.$$

This definition proves to be independent of the specific choices of the approximating sequences for f.

With respect to applications in partial differential equations, Bochner spaces are of interest on the one hand for the choices $\Omega = G$ and $X = \mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, where G is a domain in one of the Euclidean spaces $\mathbb{R}^d$ and $L^p(G, \mathbb{K})$ is nothing but the Lebesgue space $L^p(G)$. On the other hand, in the weak solution theory of evolution equations we chose Ω to be a finite or infinite interval $I \subset \mathbb{R}$, and the Banach space X may be any complete function space $\mathbb{F}$ consisting of differentiable mappings $f : G \to \mathbb{K}$, so we are dealing with the Bochner spaces $L^p(I, \mathbb{F})$ in this case.

We finally introduce so-called *Bochner-Sobolev spaces*, which differ by the property that their members are at the same time Bochner-integrable and *weakly differentiable*. Let H be a separable Hilbert space and $L^2(a,b;H)$ be the Bochner space consisting of functions $f:(a,b)\to H$, that is, we have

$$\int_a^b \|f(t)\|^2\, dt < \infty$$

for $f \in L^2(a,b,;H)$. We endow this space with the inner product

$$\langle f,g\rangle := \int_a^b \langle f(t),g(t)\rangle\, dt, \qquad f,g \in L^2(a,b;H)$$

such that the norm is given as usual by $\|f\|_2 = \langle f,f\rangle^{1/2}$. Thus $L^2(a,b;H)$ is Hilbertian, and if (a,b) is a *finite* interval, the inclusion $L^2(a,b;H) \subset L^1(a,b;H)$ holds. We shall denote $L^2(a,b;H)$ a *Hilbert-Bochner space*.

For $f \in L^2(a,b;H)$, a further function $g:(a,b)\to H$ is said to be the *weak derivative* of f, if

$$\int_a^b g\phi\, dx = -\int_a^b f\phi'\, dx$$

holds for each $\phi \in C_c^\infty(a,b,X)$, that is, ϕ is an indefinitely differentiable function with compact support in (a,b). If such a function g exists, we denote $f' := g$ in this case, which should not be confused with the classical derivative of f. But notice that if the weak derivative of f is present, then f is also classical differentiable.

Having the notion of weak derivative available, we may introduce the *Hilbert-Bochner-Sobolev space*

$$H^1(a,b;H) := W^{1,2}(a,b,H) := \{f \in L^2(a,b;H) : f' \in L^2(a,b;H)\},$$

which is a Hilbert space as well with inner product given by

$$\langle f,g\rangle_1 := \int_a^b \langle f,g\rangle\, dt + \int_a^b \langle f',g'\rangle\, dt\,, \quad f,g \in H^1(a,b;H).$$

Finally, let H_1, H_2 be *real* Hilbert spaces, where H_1 is continuously embedded into H_2. Then there is also a continuous embedding

$$H^1(a,b;H_1) \overset{\kappa}{\hookrightarrow} H^1(a,b;H_2).$$

Consequently, given a Hilbert space triple $V \overset{\iota_1}{\hookrightarrow} H \overset{\iota_2}{\hookrightarrow} V^*$, we may regard the space $H^1(a,b;V)$ as a linear subspace of $H^1(a,b;V^*)$,

which enables us to define the *mixed Bochner-Sobolev space*

$$H^1(a,b;\,V,\,V^*) \;:=\; \{f \in L^2(a,b;\,V) : f' \in L^2(a,b;\,V^*)\}$$

or equivalently

$$H^1(a,b;\,V,\,V^*) \;:=\; L^2(a,b;\,V) \cap H^1(a,b;\,V^*).$$

Then the mapping

$$(f,g) \mapsto \langle f,g\rangle_1 \;:=\; \int_a^b \langle f(t),g(t)\rangle_V \,dt \;+\; \int_a^b \langle f'(t),g'(t)\rangle_{V^*}\,dt$$

fulfills the properties on an inner product and makes $H^1(a,b;\,V,\,V^*)$ into a Hilbert space having the following properties:

(a) Each $f \in H^1(a,b;\,V,\,V^*)$ is λ-nearly everywhere identical to a continuous function $f_c : [a,b] \to H$, and by identifying the functions f and f_c the space $H^1(a,b;\,V,\,V^*)$ can continuously be embedded into the space $C([a,b],H)$.

(b) The space $C^\infty([a,b],\,V)$ is dense in $H^1(a,b;\,V,\,V^*)$.

(c) The following product rule is valid for each $t \in (a,b)$:

$$\frac{d}{dt}\,\|u(t)\|^2 \;=\; 2\,\mathrm{Re}\,\langle u'(t),u(t)\rangle_1 \text{ for all } u \in H^1(a,b;\,V,\,V^*).$$

7. Applications

In this section let G be a *domain* in $\mathbb{R}^d$, that is, an open and connected set within an Euclidean space $\mathbb{R}^d$, which may be bounded, unbounded or even the space $\mathbb{R}^d$ itself. The application part of this section consists mainly of an enumeration of function spaces having meaning in PDE theory and their adjoints, which is followed by some examples of orthonormal bases in Hilbert space $L^2(I)$, where $I \subset \mathbb{R}$ is an interval.

Preliminary, we will work in this section with *multi-indices*, which are nothing but d-tuples $\alpha = (\alpha_1,\ldots,\alpha_d)$ with values $\alpha_i \in \mathbb{N}_0$ and modulus $|\alpha| \;:=\; \alpha_1 + \ldots + \alpha_d$. They are mostly used to shorten multiple partial derivatives $\partial^\alpha f \;:=\; \partial_1^{\alpha_1}\partial_2^{\alpha_2}\ldots\partial_d^{\alpha_d} f$ of a sufficiently often differentiable and complex-valued function f, but also to denote the multiple powers $\xi^\alpha \;:=\; \xi_1^{\alpha_1}\cdot\xi_2^{\alpha_2}\cdot\ldots\cdot\xi_d^{\alpha_d}$ of a vector $\xi = (\xi_1,\xi_2,\ldots,\xi_d) \in \mathbb{C}^d$.

1. *Spaces of continuous functions.* The space $C(\overline{G})$ consists of functions $f : G \to \mathbb{K}$, which are restrictions of continuous functions $\tilde{f} : G' \to \mathbb{K}$, where G' is an open superset of G such that $\overline{G} \subset G'$

(notation: $G \Subset G'$). It is a well-known fact that $C(\overline{G})$ is complete with respect to the *supremum norm*

$$\|f\|_\infty := \sup_{x \in G} |f(x)|.$$

This is true as well for the linear subspace $C_0(\overline{G}) \subset C(\overline{G})$, consisting of functions that are vanishing at infinity, that is, for each $\epsilon > 0$ there is a compact set $C_{f,\epsilon} \subset G$ such that $|f(x)| < \epsilon$ for all points $x \in G \setminus C_{f,\epsilon}$.

2. *Spaces of continuously differentiable functions.* The linear subspace $C^k(\overline{G}) \subset C(\overline{G})$ consists of those continuously differentiable functions $f : G \to \mathbb{K}$, whose partial derivatives $\partial^\alpha f$ are bounded on $\overline{G}$ for every multi-index α with $|\alpha| \leq k$. It is complete if endowed with the norm

$$\|f\|_k := \sum_{|\alpha| \leq k} \sup_{x \in \overline{G}} |\partial^\alpha f(x)|.$$

The spaces $C^k(\overline{G})$ are not reflexive.

3. *Hölder spaces.* It has been shown that the spaces $C^k(\overline{G})$ are not very convenient for the investigation of partial differential equations, but the so-called *Hölder spaces* $C^{k,\gamma}(\overline{G})$ are so. Preliminary, a function $f : G \to \mathbb{K}$ is called *Hölder-continuous* with respect to the parameter $\gamma \in (0, 1]$, if its *Hölder-norm* is finite, that is,

$$|f|_\gamma := \sup_{x,y \in G,\, x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\gamma} < \infty.$$

Let $C^{k,\gamma}(\overline{G})$ be the linear space of functions $f \in C^k(\overline{G})$, whose partial derivatives $\partial^\alpha f$ are Hölder-continuous for all $|\alpha| = k$. Then the *Hölder norm* $|f|_{k,\gamma}$ of a function $f \in C^{k,\gamma}(\overline{G})$ is defined by the sum of the norm $\|f\|_k$ and the Hölder norms of the partial derivatives $\partial^\alpha f$ for $|\alpha| = k$:

$$|f|_{k,\gamma} := \|f\|_k + \sum_{|\alpha| = k} |\partial^\alpha f|_\gamma.$$

The space $C^{k,\gamma}(\overline{G})$ is complete with respect to this norm, but it is not a reflexive one. For $k = 0$ and $\gamma = 1$ we obtain the space $C^{0,1}(\overline{G})$ of *Lipschitz-continuous* functions.

If G is a bounded domain, then we have for $0 < \alpha < \beta \leq 1$

$$C^1(\overline{G}) \subsetneq C^{0,\beta}(\overline{G}) \subsetneq C^{0,\alpha}(\overline{G}) \subsetneq C(\overline{G}).$$

4. *Spaces of smooth functions.* Next we consider three function spaces consisting of indefinitely differentiable functions $f : G \to \mathbb{K}$ that frequently occure in applied analysis and constitute Fréchet spaces respectively complete locally convex linear spaces.

   (a) The space $C^\infty(G)$ of functions $f : G \to \mathbb{K}$ being indefinitely differentiable. Unfortunately, there is no norm that makes $C^\infty(G)$ into a complete normed space, but after all it is possible to define a countable family $\mathfrak{F}$ of seminorms to make $C^\infty(G)$ into a Fréchet space. For example, $\mathfrak{F}$ can be given by the mappings

   $$p_{N,n} := \max_{x \in K_n} \{ |(\partial^\alpha \phi)(x)| : |\alpha| \leq N \} \qquad (N, n \in \mathbb{N}),$$

   where (K_n) is a sequence of compact subsets of G such that

   $$K_1 \subset K_2 \ldots \subset G \text{ for } n \in \mathbb{N}, \quad \bigcup_{n \in \mathbb{N}} K_n = G,$$

   and each K_n lies in the interior of K_{n+1} (such a sequence is called a *compact exhaustion* of G). The space $C^\infty(G)$ is often denoted $\mathcal{E}(G)$, when considered in the locally convex topology induced by the totality of seminorms $p_{N,n}$. It is complete with respect to this topology, so it forms a Fréchet space.

   (b) The subspace $C_c^\infty(G) \subset C^\infty(G)$ of smooth functions having *compact* support

   $$\operatorname{supp}(\phi) := \overline{\{ x \in G : \phi(x) \neq 0 \}},$$

   usually denoted the *space of test functions.* This function space plays an outstanding role in analysis, since it lies dense in a multiplicity of other function spaces. Unfortunately, there is neither a countable family of seminorms making $C_c^\infty(G)$ into a Fréchet space nor is $C_c^\infty(G)$ completely metrizable.

   But it is still possible to define a locally convex topology $\mathcal{T}$ on $C_c^\infty(G)$ such that $C_c^\infty(G)$ is complete in the sense of a topological linear space. Instead of saying how the open sets of this topology $\mathcal{T}$ look like, let us define $\mathcal{T}$ by means of a suitable notion of convergence, known as $\mathcal{D}$-convergence in $C_c^\infty(G)$: a sequence (ϕ_n) in $C_c^\infty(G)$ is convergent to

$\phi \in C_c^\infty(G)$, if there is a compact set $K \subset G$ such that (i) $\operatorname{supp}(\phi_n) \subset K$ for all $n \in \mathbb{N}$ and (ii) for each multi-index α the sequence $(\partial^\alpha \phi_n)$ converges to $\partial^\alpha \phi$ *uniformly* on K. The space $C_c^\infty(G)$ equipped with the topology induced by $\mathcal{D}$-convergence is usually denoted $\mathcal{D}(G)$. Notice that we have $\mathcal{D}(G) \hookrightarrow \mathcal{E}(G)$, that is, the space $\mathcal{D}(G)$ is continuously embedded into the space $\mathcal{E}(G)$ with respect to the locally convex topologies of these spaces.

We have $\mathcal{D}(G) \hookrightarrow \mathcal{E}(G)$, that is, the space $\mathcal{D}(G)$ is continuously embedded into the space $\mathcal{E}(G)$ with respect to their locally convex topologies.

(c) The *Schwartz space* $\mathscr{S}(\mathbb{R}^d)$, which is the space of all smooth functions $\phi \in C^\infty(\mathbb{R}^d)$ having the additional property

$$|\partial^\alpha \phi(x) \cdot x^\beta| \to 0 \ \text{ for } \ \|x\|_d \to \infty$$

and arbitrary multi-indices α, β, where $\|x\|_d$ is the Euclidean norm of $x \in \mathbb{R}^d$. The elements of $\mathscr{S}(\mathbb{R}^d)$ are called *rapidly decreasing* or briefly *Schwartz functions*. Hence a Schwartz function and all its derivatives vanish at infinity faster then the reciprocal of any polynomial. The most famous example of a Schwartz function is the Hermite function

$$\mathcal{H}_0(x) \ := \ e^{-\frac{\|x\|^2}{2}} \qquad (x \in \mathbb{R}^d).$$

Rather similar to the space $C^\infty(G)$, it is possible to find for the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ a family of seminorms as follows: define for $\phi \in C^\infty(\mathbb{R}^d)$ and multi-indices α, β

$$p_{\alpha,\beta}(\phi) \ := \ \sup_{x \in \mathbb{R}^d} |\, x^\alpha (D^\beta \phi)(x) \,|. \tag{7.1}$$

Then ϕ is a Schwartz function iff $p_{\alpha,\beta}(\phi) < \infty$ holds for all indices α, β. The collection of mappings $p_{\alpha,\beta} : \mathscr{S}(\mathbb{R}^d) \to \mathbb{R}$ defines a countable family of seminorms, which gives rise to a locally convex topology $\mathcal{T}_{\mathscr{S}}$ on $\mathscr{S}(\mathbb{R}^d)$. Moreover, since the Schwartz space is complete with respect to $\mathcal{T}_{\mathscr{S}}$, it actually forms a *Fréchet space*.

Comparing the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ with the function spaces $\mathcal{D}(\mathbb{R}^d)$ and $\mathcal{E}(\mathbb{R}^d)$, it is easy to see that there is the chain of inclusions

$$\mathcal{D}(\mathbb{R}^d) \ \hookrightarrow \ \mathscr{S}(\mathbb{R}^d) \ \hookrightarrow \ \mathcal{E}(\mathbb{R}^d).$$

Moreover, by comparing the locally convex topologies of these spaces, it turns out that these inclusions are continuous.

5. *Lebesgue spaces.* To introduce the spaces $L^p(G)$ for $p \in [1, \infty)$, let $\mathcal{L}^p(G)$ be the $\mathbb{K}$-linear space of measurable functions $f : G \to \mathbb{K}$ such that $|f|^p$ is Lebesgue-integrable.

   We define for $f, g \in \mathcal{L}^p(G)$ the equivalence relation $f \sim g$ according to $f = g$ almost everywhere. Then the elements of the space $L^p(G)$ are the equivalence classes in $\mathcal{L}^p(G)$ with respect to $\sim$, that is, $L^p(G) := \mathcal{L}^p(G)/\sim$. The *Lebesgue norm* in $L^p(G)$ is given by

$$\|f\|_p := \left(\int_G |f|^p \, d\mu\right)^{\frac{1}{p}}.$$

   Finally, the FISCHER-RIESZ *theorem* claims that the spaces $L^p(G)$ are complete with respect to the Lebesgue norm. Moreover, they are reflexive for $p > 1$, but $L^1(G)$ is not so.

   The case $p = 2$ gives rise to a special attention: here it is possible to derive the norm of the space $L^2(G)$ from the inner product

$$\langle f, g \rangle := \int_G f \, \overline{g} \, d\mu,$$

   thus $\|f\|_2 = \langle f, f \rangle^{1/2}$ and $L^2(G)$ becomes into a Hilbert space.

   For $p = \infty$ and any measurable function $f : G \to \mathbb{K}$ we define the mapping

$$\|f\|_\infty := \inf\{\, a \geq 0 : |f| \leq a \ \mu\text{-almost everywhere} \,\}, \qquad (7.2)$$

   taking values in the set $\mathbb{R}^+ \cup \{\infty\}$. Let $\mathcal{L}^\infty(G)$ be the linear space of all measurable functions $f : G \to \mathbb{K}$ with $\|f\|_\infty < \infty$ and $L^\infty(G)$ be the linear space of all equivalence classes in $\mathcal{L}^\infty(G)$, where $f \sim g$ means $f = g$ μ-almost everywhere. Members of these spaces are called *essentially bounded.* Similar to the case $p < \infty$, $L^\infty(G)$ is complete with respect to (7.2). But notice that this space is not reflexive.

6. *Sobolev spaces.* A function $f : G \to \mathbb{C}$ is said to belong to the space $L^1_{loc}(G)$, if f is Lebesgue-integrable on each compact subset of G. This space is useful to define the concept of *weak differentiability.* A function $f \in L^1_{loc}(G)$ is said to be *weakly differentiable* with respect to a given multi-index α, if there is $g \in L^1_{loc}(G)$ such that

$$\int_G g(x)\,\phi(x)\,dx = (-1)^{|\alpha|}\int_G f(x)\,\partial^\alpha\phi(x)\,dx \text{ for } \phi \in C_c^\infty(G).$$

We denote $\partial^\alpha f := g$ in this case, and weak differentiability arises as a linear mapping $\partial^\alpha : D_\alpha \subset L^1_{loc}(G) \to L^1_{loc}(G)$, where each D_α is a linear subspace of $L^1_{loc}(G)$. Notice that the concept of weak derivative is compliant with that of the classical derivative: if f has classical derivative $\partial^\alpha f$, then f is also weakly differentiable, and the weak derivative is equal to $\partial^\alpha f$.

The next function spaces under consideration are those, whose elements are p-integrable and weakly differentiable up to order k at the same time. More precisely, for any integer k and real $p \geq 1$, the *Sobolev space* $W^{k,p}(G)$ is the set of functions f in the Lebesgue space $L^p(G)$, whose weak derivatives $\partial^\alpha f$ exist and also belong to $L^p(G)$ for all multi-indices α with $|\alpha| \leq k$, briefly written

$$W^{k,p}(G) := \{\, f \in L^p(G) : \partial^\alpha f \in L^p(G),\ |\alpha| \leq k \,\}.$$

Sobolev spaces are of course linear spaces, and especially for $k = 0$ the spaces $W^{0,p}(G)$ are nothing but the Lebesgue spaces $L^p(G)$ themselves.

For functions $f \in W^{k,p}(G)$ let the mapping $f \mapsto |f|_{k,p}$ be defined according to

$$|f|_{k,p} := \Big(\sum_{|\alpha|\leq k}\int_G |\partial^\alpha f|^p\,dx\Big)^{1/p}. \tag{7.3}$$

It fulfills all the properties of a norm, thus the space $W^{k,p}(G)$ is actually a normed space, and the mapping defined by (7.3) is usually denoted a *Sobolev norm*. With respect to the question of completeness, it is not hard to show that the spaces $W^{k,p}(G)$ are complete and therefore constitute Banach spaces as well.

There is a second way to define Sobolev space. Consider the subspace

$$C^\infty_{k,p}(G) := \{\, f \in C^\infty(G) : |f|_{k,p} < \infty \,\},$$

and since this space is *not* complete with respect to the norm $|\cdot|_{k,p}$, we set

$$\mathcal{W}^{k,p}(G) := \overline{C^\infty_{k,p}(G)}.$$

The space $\mathcal{W}^{k,p}(G)$ is defined by a purely topological operation, that is, by taking the closure of a smaller space. Thus for elements

in this space there is no instrinsic notion of derivative, and one introduces the notion of *strong derivative* for $u \in \mathcal{W}^{k,p}(G)$ as follows: given a multi-index α ($|\alpha| \leq k$), a further function $\partial^\alpha u$ also belonging to $\mathcal{W}^{k,p}(G)$ is said to be the α-strong derivative of u, if there is a sequence $\{\phi_n\} \in C^\infty_{k,p}(G)$ converging to u such that

$$|\partial^\alpha u - \partial^\alpha \phi_n)|_{k,p} \to 0 \qquad (n \to \infty).$$

Sobolev spaces are reflexive for $1 < p < \infty$, but not for $p = 1$.

The MEYERS-SERRIN *theorem* claims $\mathcal{W}^{k,p}(G) \cong W^{k,p}(G)$ in the sense of Banach space isomorphisms. Moreover, the notions of strong and weak derivative are equivalent when identifying functions in $\mathcal{W}^{k,p}(G)$ and $W^{k,p}(G)$.

One of the most important properties of Sobolev spaces is given by the fact that under certain hypotheses they can continuously be embedded into further spaces and those of type $C^k(\overline{G})$. We define for fixed $1 \leq p < \infty$ the *Sobolev numbers* $\theta_{k,d,p}$ by

$$\theta_{k,d,p} := k - d/p \qquad (k, d \in \mathbb{N}, p \geq 1),$$

which play a crucial role in SOBOLEV*'s embedding theorem*:

THEOREM 7.1. Let $G \subset \mathbb{R}^d$ be a domain having Lipschitz-continuous boundary. If $k, m \in \mathbb{N}$ are chosen such that $\theta_{k,d,p} > m$, then there are continuous embeddings $W^{k,p}(G) \hookrightarrow C^m(\overline{G})$ and $W^{k,p}(G) \hookrightarrow C^{m,\gamma}(\overline{G})$ for $0 < \gamma < 1$.

Hence each equivalent class $f \in W^{k,p}(G)$ has a representative in these spaces, supposed the Sobolev number is sufficiently large.

In case of $\theta_{k,d,p} \leq 0$, however, one cannot embed $W^{k,p}(G)$ into $C(\overline{G})$. But it is at least possible to find an embedding of $W^{k,p}(G)$ into some lower-dimensional Lebesgue spaces $L^q(G)$:

THEOREM 7.2. For $G \subset \mathbb{R}^d$, $k \in \mathbb{N}$ and $p, q \geq 1$ such that

$$1 < q < \frac{dp}{d - kp},$$

there are continuous embeddings $W^{k,p}(G) \hookrightarrow L^q(G)$.

Finally, there are also embeddings of Sobolev spaces into further Sobolev spaces, where the parameter p is fixed:

THEOREM 7.3. If $G \subset \mathbb{R}^d$ has Lipschitz-continuous boundary, $s, s' \in \mathbb{N}$ and $1 < s < s' < \infty$, then $W^{s',p}(G) \hookrightarrow W^{s,p}(G)$.

7. *Hilbert-Sobolev spaces.* Let us consider in the family $W^{k,p}(G)$ of Sobolev spaces the case $p=2$ more specifically and denote the space $H^k(G) := W^{k,2}(G)$ for $k \in \mathbb{N}_0$, which is often called a *Hilbert-Sobolev space.*

   Each space $H^k(G)$ can be endowed with the inner product

$$\langle f, g \rangle_k := \sum_{|\alpha| \le k} \int_G \partial^\alpha f \, \partial^\alpha \overline{g} \, dx \,, \qquad f, g \in H^k(G)$$

   and with norm $|f|_k = \langle f, f \rangle^{1/2}$ for $f \in H^k(G)$, so $H^k(G)$ forms a Hilbert space in its own right.

8. *Sobolev spaces of functions having homogeneous boundary values.* The space $C_c^\infty(G)$ of test functions can surely be endowed with the norms $|\cdot|_{k,p}$, but this leads to uncomplete normed spaces. Hence it is nearby to introduce the spaces $W_0^{k,p}(G)$ for $k \in \mathbb{N}$ by completion of the space $C_c^\infty(G)$ with respect to the Sobolev norms:

$$W_0^{k,p}(G) := \overline{C_c^\infty(G)}_{|\cdot|_{k,p}}.$$

   The elements of these spaces can be identified with functions in $W^{k,p}(G)$. Furthermore, the spaces $W_0^{k,p}(G)$ form closed linear subspaces of the Sobolev spaces $W^{k,p}(G)$. Clearly, the spaces $H_0^k(G) := H_0^{k,2}(G)$ are Hilbert spaces with inner products and norms as defined for $H^k(G)$.

In the following we describe the adjoint spaces for a couple of topological linear spaces X arising in applied analysis by quoting a linear homeomorphism $X^* \cong Y$, where Y has to be chosen in a suitable way.

1. *The adjoint space of $L^p(G)$.* Let the two real numbers $p, q > 1$ be coupled by the relation $p^{-1} + q^{-1} = 1$. Then the duality statement $L^p(G)^* \cong L^q(G)$ for $1 < p < \infty$ is a consequence of the *Hölder inequality*, which claims that for $f \in L^p(G)$ and $g \in L^q(G)$ the product $f \cdot g$ is Lebesgue-integrable and satisfies the estimation

$$\|f \cdot g\|_1 \le \|f\|_p \cdot \|g\|_q \,.$$

   Thus, each function $f \in L^q(G)$ gives rise to a continuous linear

functional ϕ_f on the space $L^p(G)$ due to the mapping

$$\phi_f(g) := \int_G fg\,d\mu, \quad g \in L^p(G). \tag{7.4}$$

On the other hand, there are no more continuous linear functionals on $L^p(G)$ besides those defined by (7.4).

In the special case $p=1$ we obtain $L^1(G)^* \cong L^\infty(G)$.

2. *The adjoint space of $C_c(G)$.* It is well-known from measure theory that each continuous function $f: G \to \mathbb{R}$ is measurable, but f is in general not Lebesgue-integrable. If $f \in C_c(G)$, however, then f is bounded and this implies Lebesgue-integrability.

   A *Radon measure*, defined on the Borel subsets of a domain $G \subset \mathbb{R}^d$, is a positive Borel measure which takes finite values on compact subsets of G. Letting μ be a Radon measure on G, the mapping $\phi_\mu: C_c(G) \to \mathbb{R}$ defined by

$$\phi_\mu(f) := \int_G f\,d\mu$$

   forms a continuous linear functional, since

$$\int_G f\,d\mu \le \|f\|_\infty \cdot \mu(\mathrm{supp}(f)).$$

   In addition, $\phi_\mu(f) \ge 0$ holds for all $f \in C_c(G)$ with $f \ge 0$, and a linear functional fulfilling this property is called *positive*. So every Borel measure μ gives rise to a positive and continuous linear functional ϕ_μ on $C_c(G)$.

   Conversely, the question arises whether each continuous linear functional $\phi \in C_c(G)^*$ appears in this way. The answer is given by the RIESZ-MARKOV *theorem*, quoting that the representation of positive linear functionals on G can just be done by the Radon measures on G:

   THEOREM 7.4. Let ϕ be a positive and continuous linear functional on the normed space $C_c(G)$. Then there is exactly one Radon measure $\mu: \mathcal{B}_G \to [0,\infty]$ such that

$$\phi(f) := \int_G f\,d\mu, \qquad f \in C_c(G).$$

3. *The adjoint space of $C_0(G)$.* A *complex* measure $\mu: \mathcal{B}_G \to \mathbb{C}$ is said to be *Radon*, if the so-called *variation* $|\mu|$ is Radon in the sense of a positive measure on G. Notice that $|\mu|$ is always

finite. The space of complex Radon measures $\mathscr{M}_r(G)$ proves to be complete with respect to the norm $\|\mu\| := |\mu|$ for $\mu \in \mathscr{M}_r(G)$.

The following theorem of RIESZ-MARKOV type describes the adjoint space of $C_0(G)$:

THEOREM 7.5. Let ϕ be a complex-valued and continuous linear functional defined on the space $C_0(G)$. Then there is exactly one complex Radon measure μ such that

$$\phi(f) := \int_G f \, d\mu, \qquad f \in C_0(G).$$

In conclusion, $C_0(G)^* \cong \mathscr{M}_r(G)$, so $C_0(G)^*$ is norm-isomorphic to the Banach space $\mathscr{M}_r(G)$ of complex Radon measures.

4. *The adjoint space of* $C[a,b]$. We consider a further variant of the RIESZ-MARKOV theorem for a compact set $[a,b]$ in the real line. Due to the properties of the Riemann-Stieltjes integral, each function g *of bounded variation* on $[a,b]$, gives rise to a continuous linear functional F on the space $C[a,b]$ according to

$$G(f) := \int_a^b f \, dg, \qquad (f \in C[a,b]).$$

Let $BV(a,b)$ denote the linear space of all functions of bounded variation. This space can be equipped with the norm

$$\|f\|_{bv} := |f(a)| + \sup_P \sum_{i=0}^{n_P - 1} |f(x_{i+1} - f(x_i)|,$$

where the supremum is taken over all finite partitions of $[a,b]$. The space $BV(a,b)$ is actually a Banach space in this norm, which is even norm-isomorphic to the space $\mathscr{M}_r(a,b)$ of Radon measures on $[a,b]$.

Now the RIESZ-MARKOV theorem quotes that *every* continuous linear functional G on $C[a,b]$ arises as Riemann-Stieltjes integral with respect to a certain function g of bounded variation, that is, we have $C[a,b]^* \cong BV(a,b)$.

5. *The adjoint space of* $\mathcal{D}(G)$. The notion of *distribution* forms a generalization of an ordinary function $f : G \to \mathbb{K}$. The corresponding subject in functional analysis, called the *theory of distributions*, was developed by L. SCHWARTZ in the midth of the

20th century to describe ideas and concepts of theoretical physics by an exact mathematical formalism. Distribution theory can be seen as a direct application of the abstract theory of locally convex linear spaces and their continuous linear functionals.

Remember the space $\mathcal{D}(G)$ consisting of test functions and furnished with a locally convex topology, in which this space is complete but not metrizable. Then every continuous linear functional $F : \mathcal{D}(G) \to \mathbb{C}$ is called a *Schwartz distribution*. We denote $\mathcal{D}'(G) := \mathcal{D}(G)^*$ the adjoint space of $\mathcal{D}(G)$ respectively the space of Schwartz distributions on G. It is also possible to characterize a Schwartz distribution with respect to $\mathcal{D}$-convergence, when applied to test functions: a linear functional $F : C_c^\infty(G) \to \mathbb{C}$ is a Schwartz distribution iff $F(\phi_n) \to 0$ for any $\mathcal{D}$-convergent sequence (ϕ_n) converges to the zero function.

Notice that the linear functional F_f derived from a locally integrable function f by

$$F_f(\phi) := \int_G f\,\phi\,dx, \qquad \phi \in C_c^\infty(G)$$

is continuous, and this is also true for the linear functional induced by the Radon measure μ according to

$$F_\mu(\phi) := \int_G \phi\,d\mu, \qquad \phi \in C_c^\infty(G).$$

Thus locally integrable functions as well as Radon measures may be regarded as Schwartz distributions. Notice that the space $\mathcal{D}'(G)$ is indeed extremely extensive and contains nearly all functions being important in applications, amongst others the functions belonging to the well-known spaces $L^p(G)$, $L^\infty(G)$, $C(G)$, $C(\overline{G})$ and $C_0(G)$, but also measures on G, which are regular enough in some sense.

A Schwartz distribution is said to be a *regular distribution*, if it is the continuous linear functional F_f induced by a locally integrable function $f \in L^1_{loc}(G)$, otherwise it is called *singular*. Well-known examples of singular distributions are the *Dirac distributions*

$$\delta_a(\phi) := \phi(a) \quad \text{for all } \phi \in C_c^\infty(G) \qquad (a \in G).$$

6. *The adjoint space of $\mathcal{E}(G)$.* Regarding the Fréchet space $\mathcal{E}(G)$, each element $F \in \mathcal{E}'(G) := \mathcal{E}(G)^*$ is called a *distribution with compact support*. This notion is founded by the fact that for

such F there is a compact set $K_F \subset G$, called the *support* of F such that $F\phi = 0$ for all $\phi \in \mathcal{E}(G)$ having support in $G \setminus K_F$. On the other hand, each measurable function f with compact support of course gives rise to a compactly supported distribution $F_f \in \mathcal{E}'(G)$.

Since it is always possible to increase an adjoint space by decreasing the underlying base space, there is a continuous inclusion $\mathcal{E}'(G) \hookrightarrow \mathcal{D}'(G)$, which is dual to embedding $\mathcal{D}(G) \hookrightarrow \mathcal{E}(G)$. So we basically may regard each compactly supported distribution as a Schwartz distribution.

7. *The adjoint space of* $\mathscr{S}(\mathbb{R}^d)$. Similar to the spaces $\mathcal{D}(G)$, we introduce the notion of *tempered distribution* as a continuous linear functional $T : \mathscr{S}(\mathbb{R}^d) \to \mathbb{C}$, which generalizes that of bounded and slow-growing locally integrable functions. The space of tempered distributions is usually denoted $\mathscr{S}'(\mathbb{R}^d)$.

Comparing the known kinds of distributional spaces, we obtain a chain of inclusions, which is dual to that of the base spaces:

$$\mathcal{E}'(\mathbb{R}^d) \subset \mathscr{S}'(\mathbb{R}^d) \subset \mathcal{D}'(\mathbb{R}^d).$$

IF we endow each of these spaces with the *weak*-topology*, then these embeddings prove to be continuous. Hence a distribution having compact support can also be regarded as a tempered distribution, which in turn gives rise to a Schwartz distribution.

8. *The adjoint space of* $H_0^k(G)$. The Sobolev space *with negative index* $H^{-k}(G)$ is defined as the adjoint space $H^{-k}(G) := H_0^k(G)^*$.

Notice that not all elements of $H^{-k}(G)$ can be represented by regular functions $f : G \to \mathbb{K}$, but all belong to the space $\mathcal{D}'(G)$ of Schwartz distributions. In case of $k = 1$, each $f \in H^{-1}(G)$ is of the form

$$f = f_0 + \partial_1 f_1 + \ldots + \partial_d f_d,$$

where $f_0, f_1, \ldots, f_d \in L^2(G)$. There are similar representations for $k > 1$.

More generally, the adjoint space of $W_0^{k,p}(G)$ $(1 < p < \infty)$ is the space $W^{-k,q}(G)$, where $p^{-1} + q^{-1} = 1$. Like in the case $p = 2$, the elements of $W^{-k,q}(G)$ can be regarded as Schwartz distributions on G.

Remember that the Lebesgue spaces $L^2(X)$ are prototypes of separable Hilbert spaces, where X forms the base set of a certain

measure space $(X, \mathcal{B}_X, \mu)$.

Our last subject is to enumerate orthonormal bases in $L^2(X)$, where $X = (a, b) \subset \mathbb{R}$ is a finite or infinite open interval.

1. If $(a, b) = (-\pi, \pi)$, the *trigonometrical functions* given by the functions
$$\frac{1}{\sqrt{2\pi}},\ \frac{1}{\sqrt{\pi}}\sin(x),\ \frac{1}{\sqrt{\pi}}\cos(x),\ \frac{1}{\sqrt{\pi}}\sin(2x),\ \frac{1}{\sqrt{\pi}}\cos(2x),\ \ldots$$
form an orthonormal base in the Hilbert space $L^2((-\pi, \pi))$.

2. In case of $(a, b) = (-1, 1)$, an orthonormal base, founded on polynomials, is given by the so-called *Legendre polynomials*
$$P_n(x) := \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2 - 1)^n, \quad n \in \mathbb{N},$$
together with the constant function $P_0(x) := 1$. These functions form the only orthogonal system in $L^2(-1, 1)$ that contains polynomials of any order.

3. Let (a, b) be the half-line $(0, \infty)$. Then the sequence of functions
$$L_n^\alpha(x) := x^{\alpha/2}e^{-x/2} \cdot \mathcal{L}_n^\alpha(x), \quad n \in \mathbb{N}$$
forms an orthonormal base in the space $L^2(0, \infty)$, where $\mathcal{L}_n^\alpha$ are called the *Laguerre polynomials*, defined by the equations
$$\mathcal{L}_n^\alpha(x) := \frac{x^{-\alpha}e^x}{n!}\frac{d^n}{dx^n}(e^{-x}x^{n+\alpha}), \quad n \in \mathbb{N}.$$

4. If (a, b) is the entire set $\mathbb{R}$, an orthonormal base of $L^2(\mathbb{R})$ is given by the sequence of *Hermite functions*
$$\mathcal{H}_n(x) := e^{-\frac{x^2}{2}}H_n(x), \quad n \in \mathbb{N},$$
where H_n are the *Hermite polynomials*
$$H_n(x) := (-1)^n \sum_{k_1+2k_2=n} \frac{n!}{k_1!\,k_2!}(-1)^{k_1+k_2}(2x)^{k_1}, \quad n \in \mathbb{N}.$$

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